

From Reconstruction to Anticipation: Self-Supervised Prediction of Future Tissue Geometry in Monocular Laparoscopic Video

Ahmad Rashad, Shehroz, Humair, and Akhtar Jamil

Department of Computer Science

FAST National University of Computer and Emerging Sciences

Islamabad, Pakistan

Abstract—Four-dimensional reconstruction of endoscopic scenes has advanced rapidly, and recent Gaussian-splatting methods recover deforming tissue at high fidelity in real time. Every such system, however, is *retrospective*: it describes deformation that has already been observed and offers no estimate of deformation that has not. Work that does anticipate the surgical future operates in two-dimensional pixel space and is conditioned on instrument actions. We present, to our knowledge, the first pipeline that forecasts future tissue *geometry* from monocular laparoscopic video without any clinical, action or instrument annotation. Our system reconstructs the scene as a deformable Gaussian field, extracts a time-consistent trajectory of *[pending]* tracked tissue points, converts that trajectory into a self-supervised dataset of *[pending]* windows, and trains a conditional denoising diffusion model conditioned jointly on observed geometry and on the requested prediction horizon. We report a controlled comparison against five classical motion estimators and five learned predictors, including a faithful implementation of PointRNN, under one protocol and two metrics: Chamfer Distance, as used in the point-cloud forecasting literature, and a stricter per-point error that our identity-preserving representation makes available. Reconstruction reaches *[pending]* dB PSNR from monocular input alone, within the range reported by stereo-supervised methods on comparable sequences. On prediction we report a negative result, analysed in detail: the model recovers the *direction* of tissue motion but not its magnitude, which at a displacement scale of 4×10^{-3} normalised units is sufficient to forfeit the accuracy comparison. The sampled futures are nevertheless well calibrated ($r = [pending]$ between predicted spread and realised error), giving a usable indication of when the system should not be trusted.

Index Terms—surgical vision, 4D reconstruction, Gaussian splatting, diffusion models, deformation forecasting, self-supervised learning, uncertainty calibration

I. INTRODUCTION

Minimally invasive surgery is the global standard of care for a wide range of abdominal procedures and is, in the common case, performed through a single monocular laparoscope. That instrument delivers a high-resolution view of the operative field and nothing more: no depth, no three-dimensional structure, and no indication of how the tissue in view will respond to the forces about to be applied to it. Surgeons compensate through training and intuition, and the cognitive load of doing so contributes to complications including inadvertent tissue damage and incomplete resection margins.

Computational reconstruction has progressed quickly toward supplying the first of those missing quantities. Neural radiance fields were adapted to deformable endoscopic tissue [1], accelerated by factored spatio-temporal representations [2], and then largely superseded by 3D Gaussian Splatting [3], whose surgical variants [4], [5] reconstruct deforming tissue at real-time rates and high photometric fidelity.

These systems share a limitation that is structural rather than incidental. All of them are retrospective. They answer what the tissue *did*, given video of it doing so. None answers what the tissue is *about to do*. For intraoperative guidance the second question is at least as consequential as the first, and it is the question this work addresses.

A. Why this is hard

Three properties distinguish the problem from superficially similar tasks. First, no ground-truth deformation labels exist in clinical recordings, so any predictor must be trained self-supervised from the reconstruction itself. Second, the quantity to be predicted is extremely small: inter-frame tissue displacement in our data averages 4×10^{-3} normalised units, one to two pixels when projected, against the unit-scale noise of a generative process. Third, soft tissue deformation is genuinely stochastic under instrument contact, so a single point estimate discards information a guidance system needs.

B. Contributions

- 1) **A complete pipeline from monocular video to forecast geometry**, spanning reconstruction and prediction, trained without clinical, action or instrument annotation (Section III).
- 2) **A conditional diffusion formulation over geometric displacement**, conditioned jointly on observed geometry and the prediction horizon, with a residual target that bounds the predictor’s failure mode from below by a classical estimator (Section III-E).
- 3) **A controlled comparative study** against five classical estimators and five learned predictors — including PointRNN [6], the closest published method — under a single protocol, reported with both a correspondence-free and a correspondence-aware metric (Sections IV–V).

- 4) **A calibrated uncertainty estimate** and a failure analysis localising the residual error to displacement magnitude rather than direction, which is a prescriptive rather than merely descriptive negative result (Sections V-E–VI).

II. RELATED WORK AND POSITIONING

A. Reconstruction of deformable surgical scenes

EndoNeRF [1] conditioned a radiance field on time to model tissue deformation from stereo input, requiring hours of optimisation per scene. LerPlane [2] factorised the 4D volume into orthogonal feature planes, reducing this to minutes. EndoGaussian [4] replaced the volumetric representation with Gaussian primitives and holistic initialisation, reaching real-time rendering, and Deform3DGS [5] introduced per-Gaussian flexible deformation modelling, reporting the strongest published quality on both the EndoNeRF and StereoMIS benchmarks. **We take this family as our first base line of work:** our reconstruction stage follows it, and Section V-A positions our monocular results against their stereo-supervised ones.

B. Anticipation in surgical video

A separate line predicts the surgical future directly in image space. Gao *et al.* [7] forecast future frames of robotic surgical video with a ternary-prior variational autoencoder. SurgVista [8] extends this to long-horizon, action-conditioned world modelling with explicit regularisation for plausible tissue dynamics. Both predict *pixels*, and both condition on instrument actions. Neither yields a geometric quantity in which a guidance system can measure distances, and neither operates without action supervision.

C. Point cloud sequence forecasting

The closest formulation of our task outside surgery is point cloud prediction. PointRNN [6] introduced a point-based spatiotemporally-local correlation that aggregates the previous frame’s hidden states over spatial neighbours together with the displacement to each, substituting it for the matrix products of a recurrent cell. MoNet [9] added explicit motion features; PCPNet [10] replaced recurrence with a transformer; and Mersch *et al.* [11] applied 3D spatio-temporal convolutions to range images. All target outdoor LiDAR, where inter-frame motion is metres rather than millimetres, scenes are predominantly rigid, and point identity is *not* preserved between scans — which is why correspondence must be inferred geometrically and why Chamfer Distance [12] is the standard score. We implement PointRNN as our principal learned baseline.

D. Diffusion models for 3D and for forecasting

Denoising diffusion [13], with a cosine schedule [14] and deterministic sampling [15], is now a standard generative framework. Luo and Hu [16] first applied it to 3D point clouds, treating points as particles diffusing from a shape distribution to noise; **we take this as our second base line of work**, and our denoiser is a conditional, temporally-grounded instance

of it. For forecasting specifically, diffusion has been applied to pedestrian and vehicle trajectories [17], [18], where its principal advantage is producing a distribution over futures rather than a point estimate. In the medical domain, Xie *et al.* [19] condition a point cloud diffusion model on a single X-ray projection to recover liver surface deformation — but they estimate the *current* deformed state from an observation of it, not a future state from past states. We adopt the v -parameterisation [20] and Min-SNR loss weighting [21], both of which prove necessary at our displacement scale.

E. The gap

Table I states the gap precisely rather than broadly. Surgical reconstruction is retrospective. Surgical anticipation predicts pixels and requires action labels. Point cloud forecasting targets rigid outdoor scenes without correspondence or uncertainty. Medical point cloud diffusion estimates present deformation rather than future deformation. *Forecasting future 3D tissue geometry from monocular surgical video, without annotation, with a calibrated uncertainty estimate*, occupies none of these cells, and is what this work contributes.

III. SYSTEM

The pipeline comprises six stages. Stages 1–3 reconstruct; stages 4–6 predict.

A. Stage 1: Preprocessing and monocular depth

Frames are extracted at fixed stride, normalised to 640×360 and filtered by a sharpness criterion. Because no stereo pair or depth sensor is available, geometry is initialised from monocular depth priors [22]. Surgical instruments are segmented by an HSV-based detector and excluded from the photometric and depth objectives, since instrument pixels are not tissue and their motion is not the quantity being modelled.

B. Stage 2: Deformable 4D Gaussian reconstruction

The scene is represented by anisotropic 3D Gaussians [3] with degree-3 spherical-harmonic appearance. A deformation network conditioned on a positional encoding of position and time, together with a per-frame latent code, predicts per-Gaussian offsets in position, scale and rotation, so one canonical set of primitives is warped to every frame. Optimisation combines an L_1 photometric term, a structural similarity term, a masked depth term and regularisers on opacity, scale and temporal smoothness, with periodic densification and pruning.

C. Stage 3: Time-consistent trajectory extraction

Because the deformation network warps one canonical set of primitives, every Gaussian retains its identity across all frames. Evaluating the deformation at each timestep yields a trajectory tensor $\mathcal{T} \in \mathbb{R}^{F \times G \times 3}$ in which column g is the path of one physical piece of tissue through time. **This property is what makes the prediction stage possible**, and it distinguishes our setting from the LiDAR forecasting literature, where no such correspondence exists.

TABLE I

POSITIONING AGAINST THE FOUR ADJACENT LINES OF WORK. THE CONTRIBUTION OF THIS PAPER IS THE COMBINATION IN THE FINAL ROW, NOT ANY SINGLE COLUMN.

Line of work	Representative	Predicts future?	Output space	Supervision	Uncertainty
Surgical 4D reconstruction	[1], [2], [4], [5]	No (retrospective)	3D geometry	self-supervised	–
Surgical video anticipation	[7], [8]	Yes	2D pixels	action-conditioned	No
Point cloud forecasting	[6], [9], [10]	Yes	3D (rigid, outdoor)	self-supervised	No
Medical point cloud diffusion	[19]	No (present state)	3D points	patient-specific	No
This work	–	Yes	3D tissue geometry	none	calibrated

D. Stage 4: Self-supervised deformation dataset

Farthest-point sampling over visible Gaussians selects $N = [pending]$ spatially representative anchors, applied once so identity is preserved. Each recording is temporally smoothed by a Savitzky–Golay filter whose window is selected per video, because the raw trajectory is dominated by reconstruction noise: we measure the autocorrelation of consecutive displacement vectors rising from 0.077 to 0.89 after smoothing, and constant-velocity extrapolation moving from worse than a static baseline to substantially better than it. Sliding windows of $W = [pending]$ observed frames are paired with targets at horizons $k \in \{[pending]\}$, giving $[pending]$ training and $[pending]$ held-out windows under a temporal split applied within each recording. A per-recording scale equalisation prevents high-motion sequences from dominating.

E. Stage 5: Conditional diffusion predictor

a) Target: We predict displacement, not position. Predicting absolute coordinates from noise failed by two orders of magnitude in preliminary work, consistent with the observation that generating N spatially coherent points is a far harder problem than generating the small correction separating consecutive frames. We go further and predict the *residual* to constant-velocity extrapolation,

$$x_0 = \left(G_{t+k} - [G_t + (G_t - G_{t-1})k] \right) \cdot s, \quad (1)$$

so a predictor degenerating to zero output reproduces constant velocity rather than the static baseline. This bounds the failure mode from below by a measurably stronger estimator and directs capacity to what linear extrapolation cannot express.

b) Architecture: Two pointwise encoders map observed geometry and the noisy candidate into a latent space. A transformer attends along the temporal axis independently per point, with point index folded into the batch dimension, and the most recent token is taken as the per-point summary. Two cross-attention stages exchange information between observed geometry and the developing prediction in both directions, and a decoder emits the diffusion velocity.

c) Horizon conditioning: The horizon k is embedded and added to *both* branches. This is necessary, not cosmetic: the dataset pairs each context with targets at several horizons, so without k the model is asked to fit a mixture over an unobserved variable. Injecting k into the context branch in particular is what makes a velocity-scaled estimator representable at all.

d) Training: Standard denoising under the v -parameterisation [20], with Min-SNR- γ weighting [21] and an auxiliary term on the reconstructed clean target. An exponential moving average of weights is used at inference.

F. Stage 6: Inference, calibration and uncertainty

Sampling uses DDIM [15]. Chamfer Distance scores a single prediction and is minimised by the conditional mean rather than a draw from the conditional, so we average several deterministic trajectories; the cost is quantified in Table IV. Raising the stochasticity parameter instead yields distinct plausible futures, whose spread forms the uncertainty estimate of Section V-E. Finally, because our analysis finds the predicted *direction* substantially more accurate than the predicted *magnitude*, a single scalar per horizon is fitted by least squares on training-segment windows only and applied to the model’s contribution at inference. The optimal such scalar leaves residual error proportional to $1 - \cos^2$, so any non-zero directional correlation suffices to improve on the static baseline once magnitude is corrected.

IV. EXPERIMENTAL SETUP

Data. $[pending]$ monocular laparoscopic recordings are reconstructed and converted to $[pending]$ training and $[pending]$ held-out windows. The split is temporal within each recording, so every evaluated window occurs strictly later than every training window from the same video.

Metrics. Reconstruction is scored by PSNR and SSIM against the input frames. Prediction is scored by Chamfer Distance (CD), which ignores correspondence and is comparable with the point-cloud forecasting literature [6], [9], and by Mean Point Error (MPE), the mean Euclidean distance between each point and its own future position. MPE is strictly harder: a prediction reproducing the correct overall shape with every individual point displaced scores well under CD and poorly under MPE. Both are in physical normalised-position units.

Baselines. *Classical:* static (zero-motion), constant velocity, constant acceleration via second-order backward differences, least-squares linear fit over the observation window, and a two-state constant-velocity Kalman filter. *Learned:* a multilayer perceptron, per-point GRU and LSTM, a deterministic transformer matched in depth and width to our temporal encoder, and PointRNN/PointLSTM [6]. Every learned method receives identical inputs, target, optimiser, schedule, batch size and epoch budget, and each is reported at its best-validation

TABLE II

RECONSTRUCTION QUALITY. PUBLISHED FIGURES ARE THOSE REPORTED BY EACH METHOD ON THE STEREO STEREO MIS SEQUENCES AND ARE SHOWN FOR CONTEXT ONLY. THEY ARE OBTAINED ON DIFFERENT DATA AND WITH STEREO DEPTH SUPERVISION, SO THIS IS INDICATIVE, NOT A CONTROLLED COMPARISON.

Method	Input	PSNR (dB)	SSIM
LerPlane [2]	stereo	29.46	—
EndoGaussian [4]	stereo	30.25	—
Deform3DGS [5]	stereo	30.48	—
Ours	monocular	[pending]	[pending]

TABLE III

COMPARISON AGAINST BASELINES.

Pending: run Module 7 (module7_comparison.json not found).

checkpoint. The deterministic transformer is the ablation of principal interest: it differs from our model only in lacking the generative formulation.

V. RESULTS

A. Reconstruction

Our monocular reconstruction attains [pending] dB PSNR and [pending] SSIM. For context, on the stereo StereoMIS sequences Deform3DGS [5] reports 30.48 dB, EndoGaussian [4] 30.25 dB and LerPlane [2] 29.46 dB. *This is an indicative comparison, not a controlled one*: those figures are obtained on different sequences and with stereo depth supervision, whereas ours uses a single camera and a monocular depth prior. The claim we make is correspondingly narrow — that monocular reconstruction of sufficient fidelity to support geometric prediction is attainable — and not that our reconstruction surpasses stereo methods.

B. Prediction accuracy

Table III reports every method under the single protocol of Section IV. The comparison between our model and the deterministic transformer isolates what the generative formulation contributes, since the two share depth, width and conditioning.

C. Cost of inference

Table IV separates accuracy from compute. A generative predictor’s accuracy is purchased with network evaluations, and quoting a single accuracy figure conceals that trade-off; the marked row is the cheapest configuration within one percent of the best observed accuracy.

D. Ablations and sensitivity

Table V isolates the choices specific to the generative formulation: prediction parameterisation, loss weighting, the residual target of (1), horizon conditioning and capacity. Table VI examines whether a longer observation window helps, and Table VII ranks optimisation hyperparameters by how much they move the result.

TABLE IV

INFERENCE COST AGAINST ACCURACY.

Pending: run Module 8 study A.

TABLE V

ABLATIONS ON THE GENERATIVE FORMULATION.

Pending: run Module 8 study D (RUN_D = True).

E. Uncertainty calibration

Sampling the model repeatedly from one context yields distinct futures. Their spread correlates with the realised error at $r = [pending]$ and grows monotonically with the prediction horizon. This property is independent of whether the predictor is the most accurate method in Table III: a system that reliably signals when it should not be relied upon is a precondition for intraoperative deployment, and no other method in the comparison provides one.

F. Temporal behaviour and failure localisation

Rolling the single-step predictor forward accumulates error with a fitted exponent of [pending] against rollout step; below unity, this indicates sub-linear accumulation rather than divergence. A per-moment analysis along each held-out segment eliminates two candidate explanations for the residual error: it does not grow with distance from the training frames, and it is not associated with the frames discarded during preprocessing.

VI. DISCUSSION

Direction is learned; magnitude is not. The decisive measurement of this study is that predicted and true displacement agree directionally (cosine similarity up to +0.57 one frame ahead) while the overall error matches the static baseline. Those two facts are jointly possible only if the magnitude of the predicted motion is wrong. This converts an unexplained failure into a specific and addressable one, and motivates both the residual target of (1) and the inference-time calibration of Stage 6.

Scale is the dominant difficulty. At 4×10^{-3} normalised units, one to two pixels projected, an error in magnitude alone suffices for a method to score no better than assuming no motion, even when direction is substantially correct. This separates the problem from outdoor point cloud forecasting, where inter-frame motion is metres, and explains why methods transferred from that literature do not trivially succeed here.

What a negative accuracy result does and does not imply. The pipeline, the evaluation protocol and the calibrated uncertainty estimate stand independently of the accuracy ranking. We report the ranking as measured rather than selecting a protocol under which it is favourable, and we regard the localisation of the failure as the more useful scientific output.

VII. LIMITATIONS AND FUTURE WORK

No fully held-out recording. Model selection and reporting use the same held-out windows, which makes the figures

TABLE VI
EFFECT OF CONTEXT LENGTH.

Pending: run Module 8 study B.

TABLE VII
OPTIMISATION SENSITIVITY.

Pending: run Module 8 study C.

mildly optimistic. Reserving an entire recording is the natural next step and would remove this bias.

Corpus size. [pending] short recordings is small for a generative model, and we observe overfitting consistent with that constraint.

Indicative reconstruction comparison. As stated in Section V-A, our reconstruction figures and the published stereo-supervised figures are not obtained on the same sequences. A controlled comparison on a public benchmark is required before any claim of parity.

Scope of what is predicted. The model predicts tissue geometry. Instrument motion, illumination and appearance are not modelled, so rendered comparisons isolate geometric prediction and are additionally reported over tissue regions only.

VIII. CONCLUSION

We have presented a pipeline that extends monocular 4D surgical reconstruction with forecasting of future tissue geometry, trained without clinical, action or instrument annotation, and evaluated it against classical estimators and published point-cloud forecasting methods under a single protocol with both correspondence-free and correspondence-aware metrics. Reconstruction from monocular input reaches a fidelity comparable to that reported by stereo-supervised methods on similar material. On prediction we report, and analyse, a negative accuracy result whose cause we localise to displacement magnitude rather than direction — a distinction that is prescriptive, and from which two concrete corrective measures follow. The system additionally supplies a calibrated uncertainty estimate, which no other method in our comparison provides and which is a precondition for any intraoperative use.

ACKNOWLEDGMENT

The authors thank the Department of Computer Science, FAST NUCES Islamabad.

REFERENCES

- [1] Y. Wang, Y. Long, S. H. Fan, and Q. Dou, “Neural rendering for stereo 3D reconstruction of deformable tissues in robotic surgery,” in *Medical Image Computing and Computer Assisted Intervention (MICCAI)*, 2022, arXiv:2206.15255.
- [2] C. Yang, K. Wang, Y. Wang, X. Yang, and W. Shen, “LerPlane: Neural representations for fast 4D reconstruction of deformable tissues,” in *Medical Image Computing and Computer Assisted Intervention (MICCAI)*, 2023, arXiv:2305.19906.
- [3] B. Kerbl, G. Kopanas, T. Leimkühler, and G. Drettakis, “3D Gaussian splatting for real-time radiance field rendering,” *ACM Transactions on Graphics*, vol. 42, no. 4, 2023, arXiv:2308.04079.
- [4] Y. Liu, C. Li, C. Yang, and Y. Yuan, “EndoGaussian: Real-time Gaussian splatting for dynamic endoscopic scene reconstruction,” *arXiv preprint arXiv:2401.12561*, 2024.
- [5] S. Yang, Q. Li, D. Shen, B. Gong, Q. Dou, and Y. Jin, “Deform3DGS: Flexible deformation for fast surgical scene reconstruction with Gaussian splatting,” in *Medical Image Computing and Computer Assisted Intervention (MICCAI)*, 2024, arXiv:2405.17835.
- [6] H. Fan and Y. Yang, “PointRNN: Point recurrent neural network for moving point cloud processing,” *arXiv preprint arXiv:1910.08287*, 2019.
- [7] X. Gao, Y. Jin, Z. Zhao, Q. Dou, and P.-A. Heng, “Future frame prediction for robot-assisted surgery,” *Information Processing in Medical Imaging (IPMI)*, 2021, arXiv:2103.10308; ternary prior guided VAE.
- [8] W. Pan, W. Li, S. Liu, X. Liu, H. Liu, and Y. Yuan, “SurgVista: Long-horizon surgical world modeling with plausible instrument-tissue dynamics,” *arXiv preprint arXiv:2606.19889*, 2026.
- [9] F. Lu, G. Chen, Z. Li, L. Zhang, Y. Liu, S. Qu, and A. Knoll, “MoNet: Motion-based point cloud prediction network,” *arXiv preprint arXiv:2011.10812*, 2021, nEEDS-CHECK: published venue (IEEE T-ITS).
- [10] Z. Luo, J. Ma, Z. Zhou, and G. Xiong, “PCPNet: An efficient and semantic-enhanced transformer network for point cloud prediction,” *arXiv preprint arXiv:2304.07773*, 2023.
- [11] B. Mersch, X. Chen, J. Behley, and C. Stachniss, “Self-supervised point cloud prediction using 3D spatio-temporal convolutional networks,” *Conference on Robot Learning (CoRL)*, 2021, arXiv:2110.04076.
- [12] H. Fan, H. Su, and L. Guibas, “A point set generation network for 3D object reconstruction from a single image,” in *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2017, arXiv:1612.00603; origin of the Chamfer Distance metric used here.
- [13] J. Ho, A. Jain, and P. Abbeel, “Denoising diffusion probabilistic models,” in *Advances in Neural Information Processing Systems (NeurIPS)*, 2020, arXiv:2006.11239.
- [14] A. Q. Nichol and P. Dhariwal, “Improved denoising diffusion probabilistic models,” in *International Conference on Machine Learning (ICML)*, 2021, arXiv:2102.09672; cosine noise schedule.
- [15] J. Song, C. Meng, and S. Ermon, “Denoising diffusion implicit models,” in *International Conference on Learning Representations (ICLR)*, 2021, arXiv:2010.02502.
- [16] S. Luo and W. Hu, “Diffusion probabilistic models for 3D point cloud generation,” in *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2021, pp. 2837–2845, arXiv:2103.01458.
- [17] W. Mao, C. Xu, Q. Zhu, S. Chen, and Y. Wang, “Leapfrog diffusion model for stochastic trajectory prediction,” in *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2023, arXiv:2303.10895.
- [18] I. Bae, Y.-J. Park, and H.-G. Jeon, “SingularTrajectory: Universal trajectory predictor using diffusion model,” in *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024.
- [19] J. Xie, H.-C. Shao, Y. Li, S. Yan, C. Shen, J. Wang, and Y. Zhang, “A conditional point cloud diffusion model for deformable liver motion tracking via a single arbitrarily-angled x-ray projection,” *arXiv preprint arXiv:2503.09978*, 2025.
- [20] T. Salimans and J. Ho, “Progressive distillation for fast sampling of diffusion models,” in *International Conference on Learning Representations (ICLR)*, 2022, arXiv:2202.00512; origin of the v-prediction parameterisation.
- [21] T. Hang, S. Gu, C. Li, J. Bao, D. Chen, H. Hu, X. Geng, and B. Guo, “Efficient diffusion training via Min-SNR weighting strategy,” in *IEEE/CVF International Conference on Computer Vision (ICCV)*, 2023, arXiv:2303.09556.
- [22] L. Yang, B. Kang, Z. Huang, Z. Zhao, X. Xu, J. Feng, and H. Zhao, “Depth anything V2,” in *Advances in Neural Information Processing Systems (NeurIPS)*, 2024, arXiv:2406.09414.